

- 19** Transition metal ions can react in a number of different ways. Listed are four different reactions of transition metal ions with relevant observations.

Which observation is due to ligand exchange only?

- A** When a blue-green solution of aqueous copper(II) chloride is warmed with copper and concentrated hydrochloric acid, a colourless solution of aqueous  $\text{CuCl}_2^-$  is made.
- B** When a blue solution of aqueous copper(II) sulfate is added to zinc powder, the solution becomes colourless as aqueous zinc(II) sulfate is made.
- C** When a purple solution of chromium(III) sulfate is warmed, a green solution containing  $[\text{Cr}(\text{H}_2\text{O})_5\text{SO}_4]^+$  is made.
- D** When a solution of aqueous iron(II) ions is added to a purple solution of  $\text{MnO}_4^-$  ions, the purple colour fades and a pale yellow solution of aqueous iron(III) ions is made.